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Every student at College X must take two and no more of the following five subjects: computing, electronics, mechanics, architecture, and psychology. The following constraints apply:

- Students taking computing cannot take psychology.
- Students taking electronics cannot take architecture.
- Students taking mechanics cannot take electronics.

Is Student G at College X taking psychology?

1. Student G is taking architecture as one of her two required subjects. This implies that she did not take electronics. Given the constraints, she would have taken psychology only if she was not taking computing. But we have no information that she is not taking computing. Therefore, (1) is not sufficient by itself to elicit a definitive answer to the question posed; NOT sufficient.
2. This provides that Student G is not taking computing. If she were, she would not be free to take psychology. Therefore, (2) is not sufficient alone to elicit a definitive answer to the question posed; NOT sufficient.

The information in (1) and (2) together with the stimulus is not sufficient to provide a definitive answer to the question. (1) tells us neither that Student G is taking psychology nor that she is not. (2) tells us that Student G is free to take psychology, but it tells us neither that she is nor is not taking it.

The correct answer is E; both statements together are still not sufficient.

•

A candy shop owner must mix three types of sweeteners—A, B, and C—in a certain ratio to get the desired mixture in his candies. What is the ratio of the weights of Sweetener A to Sweetener B to Sweetener C in the final mixture?

1. Sweeteners A, B, and C cost \$40, \$50, and \$60 per pound. The average (arithmetic mean) cost of the final mixture is \$50 per pound.
2. In the final mixture, the average (arithmetic mean) price of Sweeteners A and B together is \$45, and the average (arithmetic mean) price of Sweeteners B and C together is \$55.

Applied Problems

Three sweeteners A, B, and C are mixed as ingredients for candy. We are asked to determine the ratio among the weights of each based on information about cost per pound for each of the three ingredients and information about average (arithmetic mean) cost per pound for mixtures of two or three of the ingredients.

1. This information will not allow us to calculate weights of individual ingredients in the mixture, or even a ratio between individual weights. Therefore, (1) is not sufficient by itself; NOT sufficient.
2. This information will not allow us to calculate weights of individual ingredients in the mixture or even a ratio between the weights of individual ingredients. Therefore, (2) is not sufficient by itself; NOT sufficient.

If we combine the information provided in (1) and (2), we will be able to calculate the ratio between the weights of each of the three ingredients in the mixture.

Let a be the weight in pounds of Sweetener A, b the weight in pounds of Sweetener B, and c the weight in pounds of Sweetener C. From (1) and (2), we get the following equations:

$(40a + 50b) = 45(a + b)$

$(50b + 60c) = 55(b + c)$.

From the first, we can infer $5b - 5a = 0$, and therefore that $a = b$. Similarly, from the second, we can infer that $b = c$. So $a : b : c = 1 : 1 : 1$. So (1) and (2) are sufficient together.

The correct answer is C; both statements together are sufficient.

•

A movie theater increased the ticket price for evening shows by 10% and decreased the ticket price for morning shows by 40%. The average number of tickets sold for the morning shows increased by 30% and the average number of tickets sold for the evening shows decreased by 5%. What was the percentage change in total revenue?

1. Prior to the changes in ticket prices, the ticket prices for the morning shows and the evening shows were the same.

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same.

2. Prior to the changes in ticket prices, the ratio between the number of tickets sold for the morning shows and the evening shows was 5:8.

Applied Problems

Two changes are made in a theater's pricing plan for its morning and evening shows. The new plan raises the cost of a ticket for an evening show by 10% and reduces the cost of a ticket for a morning show by 40%. Average attendances at morning shows increased by 30%, and average attendances at evening shows decreased by 5%. These changes affected total revenue. We are asked to determine the net change in revenue that resulted from these changes.

1. This provides no information that would tell us, either explicitly or by inference, about the relative sizes of attendances at the morning and evening shows prior to the changes in pricing. To calculate the percentage changes in revenue, we would need such information. Therefore, (1) is not sufficient by itself; NOT sufficient.
2. This provides the ratio between total attendances at morning shows and evening shows prior to the changes in pricing, information that is lacking in (1). This information ultimately allows us to calculate a revised ratio for total attendances at morning shows and evening shows after the price changes. However, it lacks another crucial piece of information for calculating the change in total revenue, i.e., the information provided in (1). Therefore (2) is not sufficient by itself; NOT sufficient.

Although neither (1) nor (2) is sufficient alone, the information lacking in (1) is provided in (2), and the information lacking in (2) is provided in (1). Therefore, by combining the information in (1) and (2), the needed calculation can be made.

The correct answer is C; both statements together are sufficient.

•

George is an employee at a firm. For each day that he is absent, he forfeits 4% of his monthly salary. For every hour of overtime he does, he receives 1% added to his monthly salary. His salary reduced for the month of June by 5%. How many hours of overtime did George do in that month?

1. George was absent for 3 days in the month of June.
2. George received 7% of his monthly salary for doing overtime in the month of June.

Applied Problems

For each unpaid absence from work in a given month, an employee, George, loses 4% of his monthly salary. However, for each hour of overtime worked in that month, he can earn an addition of 1% to his salary, which helps to defray some of the salary losses due to absences. In June his monthly salary was reduced by 5%. We need to determine how many overtime hours George worked in June.

1. George's absence for 3 days in June would have incurred a salary reduction of 12%. But his total salary reduction was only 5%. The only other way for George to have reduced his net salary loss was by doing overtime. Therefore, George did 7 hours of overtime to defray 7% of the 12% salary loss resulting from unpaid absences. So (1) is sufficient by itself to provide a definitive answer to the question asked; SUFFICIENT.
2. Given that each hour of overtime earns an addition of 1% to monthly salary, George must have worked 7 hours of overtime in June. So (2) is sufficient alone; SUFFICIENT.

The correct answer is D; each statement alone is sufficient.

Questions 363 to 409 — Difficulty: Hard

363. Did the population of Town C increase by at least 100 percent from the year 2000 to the year 2010?

1. The population of Town C in 2000 was $\frac{2}{3}$ of the population in 2005.
2. The population of Town C increased by a greater number of people from 2005 to 2010 than it did from 2000 to 2005.

Algebra Percents

Determine whether the population of Town C increased by at least 100 percent from the year 2000 to the year 2010.

1. Given that the population of Town C in 2000 was $\frac{2}{}$ of the population in 2005, it is not possible to

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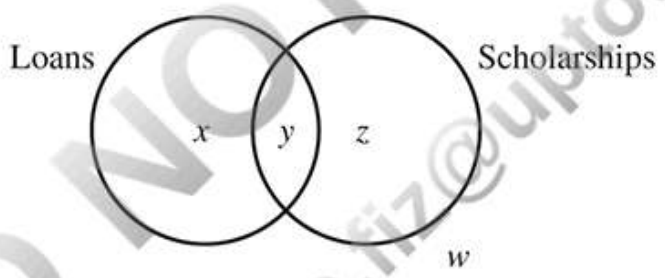
1. Given that the population of Town C in 2000 was $\frac{2}{3}$ of the population in 2005, it is not possible to determine whether the population of Town C increased by at least 100 percent from the year 2000 to the year 2010 because no information is given that helps to determine the population in 2010; NOT sufficient.
2. Given that the population of Town C increased by a greater number of people from 2005 to 2010 than it did from 2000 to 2005, it is not possible to determine whether the population of Town C increased by at least 100 percent from the year 2000 to the year 2010 because no information is given that helps determine the increase from 2000 to 2005; NOT sufficient.

Taking (1) and (2) together, from (1) the population in 2000 was $\frac{2}{3}$ of the population in 2005, so if P represents the population in 2005, then the population increase from 2000 to 2005 is $P - \frac{2}{3}P = \frac{1}{3}P$. From (2), the population increase from 2005 to 2010 was greater than $\frac{1}{3}P$, so the population increase from 2000 to 2010 was greater than $\frac{1}{3}P + \frac{1}{3}P = \frac{2}{3}P$. Thus, the percent increase from 2000 to 2010 is given by $\frac{\text{increase in population from 2000 to 2010}}{\text{population in 2000}}$, which is greater than $\frac{\frac{2}{3}P}{\frac{2}{3}P} = 1$ or 100%.

The correct answer is C; both statements together are sufficient.

364. In a survey of 200 college graduates, 30% said they had received student loans during their college careers, and 40% said they had received scholarships. What percent of those surveyed said that they had received neither student loans nor scholarships during their college careers?
    1. 25% of those surveyed said that they had received scholarships but no loans.
    2. 50% of those surveyed who said that they had received loans also said that they had received scholarships.

Arithmetic Sets



Using the variables shown on the Venn diagram above, determine the value of w. According to the information given, 30% had received student loans, so $x + y = 0.3(200) = 60$ and $x = 60 - y$. Also, 40% had received scholarships, so $y + z = 0.4(200) = 80$ and $z = 80 - y$. Then, since $x + y + z + w = 200$, $w = 200 - x - y - z = 200 - (60 - y) - y - (80 - y) = 60 + y$. Thus, if the value of y can be determined, then the value of w can be determined.

1. Since 25% received scholarships but no loans, $z = 80 - y = 0.25(200) = 50$ and $y = 30$; SUFFICIENT.
2. Since 50% of those who had received loans had also received scholarships, $0.5(x + y) = y$ and so $0.5(60) = 30 = y$; SUFFICIENT.

The correct answer is D; each statement alone is sufficient.

365. Stores L and M each sell a certain product at a different regular price. If both stores discount their regular price of the product, is the discount price at Store M less than the discount price at Store L?
    1. At Store L, the discount price is 10 percent less than the regular price; at Store M, the discount price is 15 percent less than the regular price.
    2. At Store L, the discount price is \$5 less than the regular store price; at Store M, the discount price is \$6 less than the regular price.

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Arithmetic Percents

Let L_r and L_d be the regular and discounted prices, respectively, at Store L, and let M_r and M_d be the regular and discounted prices, respectively, at Store M. Determine if $M_d < L_d$.

1. Knowing that $L_d = (1 - 0.10)L_r = 0.90L_r$ and that $M_d = (1 - 0.15)M_r = 0.85M_r$ gives no information for comparing M_d and L_d, NOT sufficient.
2. Knowing that $L_d = L_r - 5$ and that $M_d = M_r - 6$ gives no information for comparing M_d and L_d, NOT sufficient.

Taking (1) and (2) together gives $0.90L_r = L_r - 5$ and $0.85M_r = M_r - 6$, from which it follows that $0.10L_r = 5$ or $L_r = 50$ and $0.15M_r = 6$ or $M_r = 40$. Then $L_d = 50 - 5 = 45$ and $M_d = 40 - 6 = 34$. Therefore, $M_d < L_d$.

The correct answer is C; both statements together are sufficient.

366. A box contains only red chips, white chips, and blue chips. If a chip is randomly selected from the box, what is the probability that the chip will be either white or blue?

1. The probability that the chip will be blue is $\frac{1}{5}$.
2. The probability that the chip will be red is $\frac{1}{3}$.

Arithmetic Probability

1. Since the probability of drawing a blue chip is known, the probability of drawing a chip that is not blue (in other words, a red or white chip) can also be found. However, the probability of drawing a white or blue chip cannot be determined from this information; NOT sufficient.
2. The probability that the chip will be either white or blue is the same as the probability that it will NOT be red. Thus, the probability is $1 - \left(\frac{1}{3}\right) = \left(\frac{2}{3}\right)$; SUFFICIENT.

The correct answer is B; statement 2 alone is sufficient.

367. Did the sum of the prices of three shirts exceed $60?

1. The price of the most expensive of the shirts exceeded $30.
2. The price of the least expensive of the shirts exceeded $20.

Arithmetic Applied Problems

1. Given that the price of the most expensive shirt exceeded $30, the sum of the prices of the shirts can be under $60 (if the prices were $10, $10, and $35), and the sum of the prices of the shirts can be over $60 (if the prices were $10, $10, and $50); NOT sufficient.
2. Given that the price of the least expensive shirt exceeded $20, it follows that the sum of the prices of the shirts exceeds 3($20) = $60; SUFFICIENT.

The correct answer is B; statement 2 alone is sufficient.

368. What is the total number of coins that Bert and Claire have?

1. Bert has 50 percent more coins than Claire.
2. The total number of coins that Bert and Claire have is between 21 and 28.

Arithmetic Computation with Integers

Determine the total number of coins Bert and Claire have. If B represents the number of coins that Bert has and C represents the number of coins that Claire has, determine $B + C$.

1. Bert has 50% more coins than Claire, so $B = 1.5C$, and $B + C = 1.5C + C = 2.5C$, but the value of C can vary; NOT sufficient.
2. The total number of coins Bert and Claire have is between 21 and 28, so $21 < B + C < 28$ and, therefore, $B + C$ could be 22, 23, 24, 25, 26, or 27; NOT sufficient.

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therefore, $B + C$ could be 22, 23, 24, 25, 26, or 27; NOT sufficient.

Taking (1) and (2) together, $21 < 2.5C < 28$ and then $\frac{21}{2.5} < C < \frac{28}{2.5}$ or $8.4 < C < 11.2$.

If $C = 9$, then $B = (1.5)(9) = 13.5$; if $C = 10$, then $B = (1.5)(10) = 15$; and if $C = 11$, then $B = (1.5)(11) = 16.5$. Since B represents a number of coins, B is an integer. Therefore, $B = 15, C = 10$, and $B + C = 25$.

The correct answer is C; both statements together are sufficient.

369. A telephone station has x processors, each of which can process a maximum of y calls at any particular time, where x and y are positive integers. If 500 calls are sent to the station at a particular time, can the station process all of the calls?

1. $x = 600$
2. $100 < y < 200$

Algebra Applied Problems

At a particular time, the telephone station can process a maximum of xy calls, where x and y are positive integers. Determine whether $xy \geq 500$.

1. Given that $x = 600$, it follows that $xy \geq 600$ since $y \geq 1$ (y is a positive integer); SUFFICIENT.
2. Given that $100 < y < 200$, $xy < 500$ is possible (if $x = 3$ and $y = 150$), and $xy \geq 500$ is possible (if $x = 10$ and $y = 150$); NOT sufficient.

The correct answer is A; statement 1 alone is sufficient.

	Price per Flower
Roses	\$1.00
Daisies	\$0.50

370. Kim and Sue each bought some roses and some daisies at the prices shown above. If Kim bought the same total number of roses and daisies as Sue, was the price of Kim's purchase of roses and daisies higher than the price of Sue's purchase of roses and daisies?

1. Kim bought twice as many daisies as roses.
2. Kim bought 4 more roses than Sue bought.

Algebra Applied Problems

Let R_K be the number of roses that Kim bought, let R_S be the number of roses that Sue bought, and let T be the total number of roses and daisies each bought. Then Kim bought $(T - R_K)$ daisies and Sue bought $(T - R_S)$ daisies. For the roses and daisies, Kim paid a total of $\$[R_K + \frac{1}{2}(T - R_K)] = \$\frac{1}{2}(R_K + T)$ and Sue paid a total of $\$[R_s + \frac{1}{2}(T - R_s)] = \$\frac{1}{2}(R_s + T)$. Determine whether $\frac{1}{2}(R_K + T) > \frac{1}{2}(R_S + T)$, or equivalently, determine whether $R_K > R_s$.

1. Given that $T - R_K = 2R_K$, or $T = 3R_K$, it is not possible to determine whether $R_K > R_S$ because no information is provided about the value of R_S; NOT sufficient.
2. Given that $R_K = R_S + 4$, it follows that $R_K > R_S$; SUFFICIENT.

The correct answer is B; statement 2 alone is sufficient.

371. Jazz and blues recordings accounted for 6 percent of the \$840 million revenue from the sales of recordings in Country Y in 2000. What was the revenue from the sales of jazz and blues recordings in Country Y in 1998?

1. Jazz and blues recordings accounted for 5 percent of the revenue from the sales of recordings in Country Y in 1998.
2. The revenue from the sales of jazz and blues recordings in Country Y increased by 40 percent from 1998 to

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2000.

Arithmetic Percents

It is given that jazz and blues recordings accounted for 6% of the $840 million revenue of recordings in Country Y in 2000. Determine the revenue from the sales of jazz and blues recordings in 1998.

1. This indicates that jazz and blues recordings accounted for 5% of the revenue from the sales of recordings in Country Y in 1998. However, no information is given to indicate what the revenue from the sales of recordings was in 1998. Therefore, the revenue from sales of jazz and blues recordings in 1998 cannot be determined; NOT sufficient.
2. This indicates that the revenue from the sales of jazz and blues recordings in Country Y increased by 40% from 1998 to 2000. Letting R_{1998} and R_{2000} represent the revenue from the sales of jazz and blues recordings in 1998 and 2000, respectively, then $R_{2000} = 1.4R_{1998}$, and so $(0.06)(\$840 \text{ million}) = 1.4R_{1998}$, from which R_{1998} can be determined; SUFFICIENT.

The correct answer is B; statement 2 alone is sufficient.

372. On a certain nonstop trip, Marta averaged x miles per hour for 2 hours and y miles per hour for the remaining 3 hours. What was her average speed, in miles per hour, for the entire trip?

1. $2x + 3y = 280$
2. $y = x + 10$

Algebra Rate Problems

Marta traveled a total of $(2x + 3y)$ miles in 2 + 3 = 5 hours for an average speed of $\left(\frac{2x+3y}{5}\right)$ miles per hour. Determine the value of $\frac{2x+3y}{5}$.

1. Given that $2x + 3y = 280$, it follows that $\frac{2x+3y}{5} = \frac{280}{5}$; SUFFICIENT.
2. Given that $y = x + 10$, it follows that $\frac{2x+3y}{5} = \frac{2x+3(x+10)}{5} = x + 6$. Therefore, the value of $\frac{2x+3y}{5}$ can be 56 (if $x = 50$ and $y = 60$), and the value of $\frac{2x+3y}{5}$ can be 61 (if $x = 55$ and $y = 65$); NOT sufficient.

The correct answer is A; statement 1 alone is sufficient.

373. A group consisting of several families visited an amusement park where the regular admission fees were ¥5,500 for each adult and ¥4,800 for each child. Because there were at least 10 people in the group, each paid an admission fee that was 10% less than the regular admission fee. How many children were in the group?

1. The total of the admission fees paid for the adults in the group was ¥29,700.
2. The total of the admission fees paid for the children in the group was ¥4,860 more than the total of the admission fees paid for the adults in the group.

Arithmetic Simultaneous Equations

Determine the number of children in a group of at least 10 people who visited an amusement park.

1. This indicates that $(0.9)(5{,}500)A = 29{,}700$, where A represents the number of adults in the group. From this, the number of adults can be determined. However, the number of children in the group cannot be determined without additional information about the exact number of people in the group; NOT sufficient.
2. If C and A represent the numbers of children and adults, respectively, in the group, this indicates that $(0.9)(4{,}800)C = (0.9)(5{,}500)A + 4{,}860$, or $48C = 55A + 54$, or $48C - 55A = 54$, which is a single equation with two variables from which unique values of C and A cannot be determined, even under the assumptions that C and A are integers such that $C + A \geq 10$. For example, the values of C and A could be 8 and 6, respectively, since (48)(8) – (55)(6) = 54, or the values of C and A could be 63 and 54, respectively, since (48)(63) – (55)(54) = 54; NOT sufficient.

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respectively, since $(46)(63) = (55)(54) = 54$, NOT sufficient.

Taking (1) and (2) together, it follows that $A = \dfrac{29{,}700}{(0.9)\,(5{,}500)} = 6$ and

$$C = \frac{(0.9)\,(5{,}500)\,(6) + 4{,}860}{(0.9)\,(4{,}800)} = 8$$

The correct answer is C; both statements together are sufficient.

374. What is the ratio of the average (arithmetic mean) height of students in Class X to the average height of students in Class Y?

1. The average height of the students in Class X is 120 centimeters.
2. The average height of the students in Class X and Class Y combined is 126 centimeters.

Arithmetic Statistics

1. Given that the average height of the students in Class X is 120 cm, but given no information about the average height of the students in Class Y, the desired ratio cannot be determined; NOT sufficient.
2. Given that the average height of the students in Class X and Class Y combined is 126 cm, the ratio of the averages of the individual classes cannot be determined. For example, if Class X consists of 10 students, each of whom has height 120 cm, and Class Y consists of 10 students, each of whom has height 132 cm, then the average height of the students in Class X and Class Y combined is $\dfrac{10\,(120) + 10\,(132)}{20} = 126$ cm and the ratio of the averages of the individual classes is $\dfrac{120}{132}$. However, if Class X consists of 10 students, each of whom has height 120 cm, and Class Y consists of 20 students, each of whom has height 129 cm, then the average height of the students in Class X and Class Y combined is $\dfrac{10\,(120) + 20\,(129)}{30} = 126$ cm and the ratio of the averages of the individual classes is $\dfrac{120}{129}$; NOT sufficient.

Taking (1) and (2) together is of no more help than either (1) or (2) taken separately because the same examples used to show that (2) is not sufficient include the information from (1).

The correct answer is E; both statements together are still not sufficient.

375. Max has \$125 consisting of bills each worth either \$5 or \$20. How many bills worth \$5 does Max have?

1. Max has fewer than 5 bills worth \$5 each.
2. Max has more than 5 bills worth \$20 each.

Arithmetic Equations

Letting x and y be the numbers, respectively, of \$5 bills and \$20 bills, it follows from the given information that $5x + 20y = 125$, or equivalently, $x + 4y = 25$. We are to determine the value of x.

1. Given that $x < 5$ and x is a nonnegative integer, the possible values of x must be among the integers 0, 1, 2, 3, and 4. Therefore, the possible values of $4y = 25 - x$ must be among the integers 25, 24, 23, 22, and 21. Since y is an integer, $4y$ must also be a multiple of 4, and hence it follows that $4y = 24$. Therefore, $x = 25 - 4y = 25 - 24 = 1$; SUFFICIENT.
2. Given that $y > 5$, the possible values of y must be among the integers 6, 7, 8, …. Therefore, the possible values of $x = 25 - 4y$ must be among the integers 1, −3, −7, …. Since x is a nonnegative integer and 1 is the only nonnegative integer among the integers 1, −3, −7, …, it follows that $x = 1$; SUFFICIENT.

The correct answer is D;each statement alone is sufficient.

376. At a bakery, all donuts are priced equally and all bagels are priced equally. What is the total price of 5 donuts and 3 bagels at the bakery?

1. At the bakery, the total price of 10 donuts and 6 bagels is \$12.90.
2. At the bakery, the price of a donut is \$0.15 less than the price of a bagel.

Algebra Simultaneous Equations

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Algebra Simultaneous Equations

Let x be the price, in dollars, of each donut and let y be the price, in dollars, of each bagel. Find the value of $5x + 3y$.

1. Given that $10x + 6y = 12.90$, since $5x + 3y = \frac{1}{2}\left(10x + 6y\right)$, it follows that $5x + 3y = \frac{1}{2}\left(12.90\right)$; SUFFICIENT.
2. Given that $x = y - 0.15$, then $5x + 3y = 5(y - 0.15) + 3y = 8y - 0.75$, which varies as y varies; NOT sufficient.

The correct answer is A; statement 1 alone is sufficient.

377. Last year, if Arturo spent a total of $12,000 on his mortgage payments, real estate taxes, and home insurance, how much did he spend on his real estate taxes?

1. Last year, the total amount that Arturo spent on his real estate taxes and home insurance was $33\frac{1}{3}$ percent of the amount that he spent on his mortgage payments.
2. Last year, the amount that Arturo spent on his real estate taxes was 20 percent of the total amount he spent on his mortgage payments and home insurance.

Arithmetic Applied Problems

Let M, R, and H be the amounts that Arturo spent last year on mortgage payments, real estate taxes, and home insurance, respectively. Given that $M + R + H = 12{,}000$, determine the value of R.

1. Given that $R + H = \frac{1}{3}M$ and $M + R + H = 12{,}000$, then $M + \frac{1}{3}M = 12{,}000$, or $M = 9{,}000$. However, the value of R cannot be determined, since it is possible that $R = 2{,}000$ (use $M = 9{,}000$ and $H = 1{,}000$), and it is possible that $R = 1{,}000$ (use $M = 9{,}000$ and $H = 2{,}000$); NOT sufficient.
2. Given that $R = \frac{1}{5}(M + H)$, or $5R = M + H$ and $M + R + H = 12{,}000$, which can be rewritten as $(M + H) + R = 12{,}000$, then $5R + R = 12{,}000$, or $R = 2{,}000$; SUFFICIENT.

The correct answer is B; statement 2 alone is sufficient.

378. Is the number of members of Club X greater than the number of members of Club Y?

1. Of the members of Club X, 20 percent are also members of Club Y.
2. Of the members of Club Y, 30 percent are also members of Club X.

Arithmetic Sets

1. Given that 20 percent of the members of Club X are also members of Club Y, these 20 percent could be the only members of Club Y, in which case all the members of Club Y belong to Club X and Club X would have a greater number of members than Club Y. However, these 20 percent could form only a tiny portion of the members of Club Y and Club Y would have a greater number of members than Club X; NOT sufficient.
2. Given that 30 percent of the members of Club Y are also members of Club X, these 30 percent could be the only members of Club X, in which case all the members of Club X belong to Club Y and Club Y would have a greater number of members than Club X. However, these 30 percent could form only a tiny portion of the members of Club X and Club X would have a greater number of members than Club Y; NOT sufficient.

Taking (1) and (2) together, the Venn diagram below shows the numbers of members in one or both of Club X and Club Y. For example, b is the number of members of Club X that are also members of Club Y, and b is the number of members of Club Y that are also members of Club X. We are to determine whether $a + b$, the number of members of Club X, is greater than $b + c$, the number of members of Club Y.

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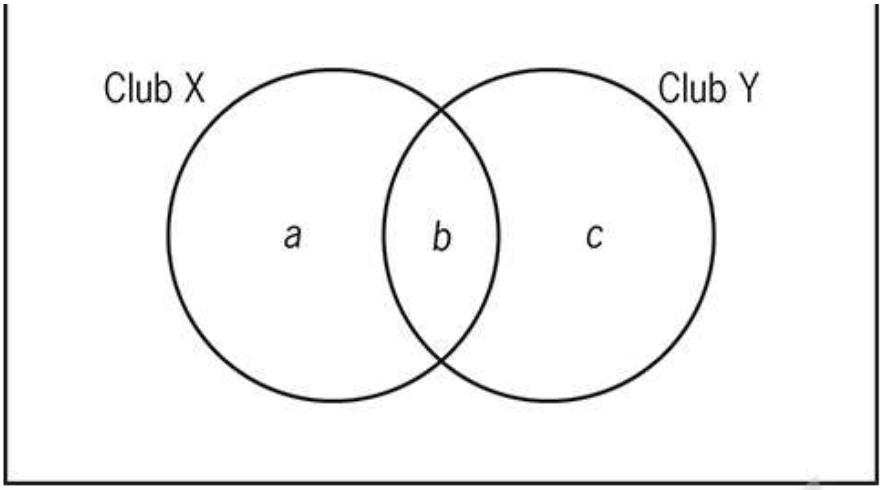

$$b = \frac{1}{5}(a+b) \quad \text{from (1) and 20\% = } \frac{1}{5}$$

$$5b = a + b \quad \text{multiply both sides by 5}$$

$$b = \frac{3}{10}(b+c) \quad \text{from (2) and 30\% = } \frac{3}{10}$$

$$\frac{10}{3}\,b = b + c \quad \text{multiply both sides by } \frac{10}{3}$$

The 2nd equation above says that the number of members of Club X is $5b$ and the 4th equation above says that the number of members of Club Y is $\frac{10}{3}b$. Since 5 is greater $\frac{10}{3}$ than, it follows that the number of members of Club X is greater than the number of members of Club Y.

The correct answer is C; both statements together are sufficient.

379. The only articles of clothing in a certain closet are shirts, dresses, and jackets. The ratio of the number of shirts to the number of dresses to the number of jackets in the closet is 9:4:5, respectively. If there are more than 7 dresses in the closet, what is the total number of articles of clothing in the closet?

1. The total number of shirts and jackets in the closet is less than 30.
2. The total number of shirts and dresses in the closet is 26.

Arithmetic Ratio and Proportion

Letting s, d, and j represent, respectively, the numbers of shirts, dresses, and jackets in the closet, then $s = 9x, d = 4x$, and $j = 5x$, where x is a positive integer. It is given that $4x > 7$, and so $4x > 8$ or $x \geq 2$ since x is an integer. Determine the value of $9x + 4x + 5x = 18x$.

1. This indicates that $9x + 5x < 30$, and so $14x \leq 28$ or $x \leq 2$ since x is an integer. It follows from $x \geq 2$ and $x \leq 2$ that $x = 2$ and $18x = 36$; SUFFICIENT.
2. This indicates that $9x + 4x = 26$ or $13x = 26$, or $x = 2$. It follows that $18x = 36$; SUFFICIENT.

The correct answer is D; each statement alone is sufficient.

380. Material A costs \$3 per kilogram, and Material B costs \$5 per kilogram. If 10 kilograms of Material K consists of x kilograms of Material A and y kilograms of Material B, is $x > y$?

1. $y > 4$
2. The cost of the 10 kilograms of Material K is less than \$40.

Algebra bInequalities

Since $x + y = 10$, the relation $x > y$ is equivalent to $x > 10 - x$, or $x > 5$.

1. The given information is consistent with $x = 5.5$ and $y = 4.5$, and the given information is also consistent with $x = y = 5$. Therefore, it is possible for $x > y$ to be true, and it is possible for $x > y$ to be false; NOT sufficient.

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NOT sufficient.

2. Given that $3x + 5y < 40$, or $3x + 5(10 - x) < 40$, then $3x - 5x < 40 - 50$. It follows that $-2x < -10$, or $x > 5$; SUFFICIENT.

The correct answer is B; statement 2 alone is sufficient.

381. At what speed was a train traveling on a trip when it had completed half of the total distance of the trip?

1. The trip was 460 miles long and took 4 hours to complete.
2. The train traveled at an average rate of 115 miles per hour on the trip.

Arithmetic Applied Problems

Determine the speed of the train when it had completed half the total distance of the trip.

1. Given that the train traveled 460 miles in 4 hours, the train could have traveled at the constant rate of 115 miles per hour for 4 hours, and thus it could have been traveling 115 miles per hour when it had completed half the total distance of the trip. However, the train could have traveled 150 miles per hour for the first 2 hours (a distance of 300 miles) and 80 miles per hour for the last 2 hours (a distance of 160 miles), and thus it could have been traveling 150 miles per hour when it had completed half the total distance of the trip; NOT sufficient.
2. Given that the train traveled at an average rate of 115 miles per hour, each of the possibilities given in the explanation for (1) could occur, since 460 miles in 4 hours gives an average speed of $\frac{460}{4} = 115$ miles per hour; NOT sufficient.

Assuming (1) and (2), each of the possibilities given in the explanation for (1) could occur. Therefore, (1) and (2) together are not sufficient.

The correct answer is E; both statements together are still not sufficient.

382. Tom, Jane, and Sue each purchased a new house. The average (arithmetic mean) price of the three houses was $120,000. What was the median price of the three houses?

1. The price of Tom's house was $110,000.
2. The price of Jane's house was $120,000.

Arithmetic Statistics

Let T, J, and S be the purchase prices for Tom's, Jane's, and Sue's new houses, respectively. Given that the average purchase price is 120,000, or $T + J + S = 3(120{,}000)$, determine the median purchase price.

1. Given $T = 110{,}000$, the median could be 120,000 (if $J = 120{,}000$ and $S = 130{,}000$) or 125,000 (if $J = 125{,}000$ and $S = 125{,}000$); NOT sufficient.
2. Given $J = 120{,}000$, the following two cases include every possibility consistent with $T + J + S = (3)(120{,}000)$, or $T + S = (2)(120{,}000)$.

   i. $T = S = 120{,}000$

   ii. One of T or S is less than 120,000 and the other is greater than 120,000.

   In each case, the median is clearly 120,000; SUFFICIENT.

The correct answer is B; statement 2 alone is sufficient.

383. A certain bookcase has 2 shelves of books. On the upper shelf, the book with the greatest number of pages has 400 pages. On the lower shelf, the book with the least number of pages has 475 pages. What is the median number of pages for all of the books on the 2 shelves?

1. There are 25 books on the upper shelf.
2. There are 24 books on the lower shelf.

Arithmetic Statistics

1. The information given says nothing about the number of books on the lower shelf. If there are fewer than 25 books on the lower shelf, then the median number of pages will be the number of pages in one of the books on the upper shelf or the average number of pages in two books on the upper shelf. Hence, the

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books on the upper shelf or the average number of pages in two books on the upper shelf. Hence, the median will be at most 400. If there are more than 25 books on the lower shelf, then the median number of pages will be the number of pages in one of the books on the lower shelf or the average number of pages in two books on the lower shelf. Hence, the median will be at least 475; NOT sufficient.

2. An analysis very similar to that used in (1) shows the information given is not sufficient to determine the median; NOT sufficient.

Given both (1) and (2), it follows that there is a total of 49 books. Therefore, the median will be the 25th book when the books are ordered by number of pages. Since the 25th book in this ordering is the book on the upper shelf with the greatest number of pages, the median is 400. Therefore, (1) and (2) together are sufficient.

The correct answer is C; both statements together are sufficient.

384. In planning for a trip, Joan estimated both the distance of the trip, in miles, and her average speed, in miles per hour. She accurately divided her estimated distance by her estimated average speed to obtain an estimate for the time, in hours, that the trip would take. Was her estimate within 0.5 hours of the actual time that the trip took?

1. Joan's estimate for the distance was within 5 miles of the actual distance.
2. Joan's estimate for her average speed was within 10 miles per hour of her actual average speed.

Arithmetic **Applied Problems; Estimating**

1. Given that the estimate for the distance was within 5 miles of the actual distance, it is not possible to determine whether the estimate of the time (that is, the estimated distance divided by the estimated average speed) was within 0.5 hour of the actual time because nothing is known about the estimated average speed; NOT sufficient.
2.

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